WEEK 2: DATA CLEANING AND QUALITY REPORT

1. Introduction

The objective of this phase was to improve the quality, consistency, and usability of the dataset collected during Week 1. Data cleaning was performed to remove irrelevant information, handle missing values, standardize data formats, and create new features that support meaningful analysis and visualization.

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2. Data Cleaning Process

2.1 Removal of Empty Columns

A total of **38 columns** containing 100% missing values were removed because they provided no analytical value and would only increase dataset complexity.

2.2 Removal of Identifier Columns

Columns such as Reference IDs and other unique identifiers were removed since they do not contribute to trend analysis or statistical evaluation.

2.3 Removal of Timestamp Columns

System-generated timestamp columns were removed because they were not relevant to the project objectives and did not provide useful business insights.

2.4 Removal of Out-of-Scope Columns

Several columns containing personal information, sensitive data, or excessive missing values were removed to improve data quality and maintain focus on the project objectives.

2.5 Handling Missing Values

- 11 records with missing Country values were removed.
- 3 records with missing Intake values were removed because Intake was required for feature engineering.

2.6 Feature Engineering

The Intake column contained multiple pieces of information in a single field. Therefore, it was split into:

- Education Level
- Season
- Month & Year
- Course

This transformation improved analytical flexibility and enabled detailed visualization.

2.7 Standardization

The text "(Fully Online)" was removed from College_1st_Choice values to ensure consistency.

2.8 Missing Value Imputation

Missing values in College_1st_Choice were replaced with "Unknown" to retain records while clearly identifying unavailable information.

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3. Data Cleaning Summary

Metric	Before Cleaning	After Cleaning
Rows	7543	7529
Columns	80	21
Fully Empty Columns	38	0
Rows Removed	14	14
Derived Columns Created	0	4

Key Outcomes

✓ Reduced dataset complexity and removed irrelevant attributes

✓ Improved data consistency and handled missing values

✓ Created new analytical features and prepared dataset for visualization and reporting

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4. Cleaned Dataset

The final cleaned dataset contains 7,529 records across 21 columns with clear variable names and standardized values.

Degree_type	Country	Continent	University	Intake_Major	Intake_Degree_Type	Intake_Yea	Intake_Season	Intake_Month
Bachelor's Degree	India	Asia	DePaul University	Human Resources	MS	2024	Fall	September '24
Bachelor's Degree	India	Asia	DePaul University	Artificial Intelligence	MS	2025	Fall	September '25
Bachelor's Degree	Pakistan	Asia	DePaul University	Biological Sciences	MS	2024	Fall	September '24
Bachelor's Degree	Pakistan	Asia	DePaul University	Biological Sciences	MS	2024	Fall	September '24
Bachelor's Degree	India	Asia	DePaul University	Supply Chain Management	MS	2025	Fall	September '25
No Degree	Thailand	Asia	DePaul University	Artificial Intelligence	MS	2024	Summer	June '24
Bachelor's Degree	Vietnam	Asia	DePaul University	Business Analytics	MS	2024	Fall	September '24
Bachelor's Degree	India	Asia	DePaul University	Information Systems	MS	2024	Winter	January '24
Bachelor's Degree	India	Asia	DePaul University	Data Science	MS	2024	Fall	September '24
Bachelor's Degree	India	Asia	DePaul University	Data Science	MS	2024	Fall	September '24
Bachelor's Degree	India	Asia	DePaul University	Data Science	MS	2024	Fall	September '24
Bachelor's Degree	India	Asia	DePaul University	Data Science	MS	2024	Fall	September '24
Bachelor's Degree	India	Asia	DePaul University	Data Science	MS	2024	Fall	September '24
Bachelor's Degree	United States	North America	DePaul University	Finance	MS	2024	Winter	January '24
Bachelor's Degree	Taiwan	Asia	DePaul University	Human_Computer Interaction	MS	2025	Fall	September '25
Bachelor's Degree	India	Asia	DePaul University	Business Analytics	MS	2023	Fall	September '23
Bachelor's Degree	India	Asia	DePaul University	Health Informatics	MS	2024	Winter	January '24
Bachelor's Degree	India	Asia	DePaul University	Computer Science	MBA	2023	Fall	September '23
Bachelor's Degree	Thailand	Asia	DePaul University	Marketing Analysis	MS	2023	Fall	September '23
Bachelor's Degree	United States	North America	DePaul University	Business Analytics	MS	2024	Fall	September '24
Bachelor's Degree	India	Asia	DePaul University	Game Programming	MS	2024	Fall	September '24

Fig 1. 1: A screenshot of the cleaned dataset table

5. Visualization and Insights

5.1 Visualization Before Cleaning

Prior to cleaning, the Intake column contained a single combined text field that merged course, education level, season, and intake period into one unanalyzable value as shown in the sample below.

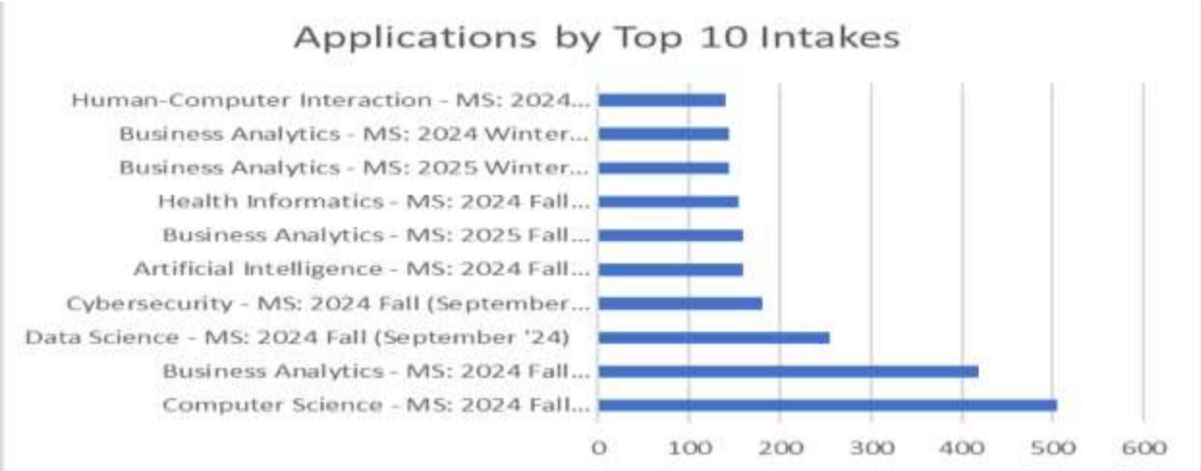

Fig 1.2 Applications of 10 Intakes (before data cleaning).

5.2 Visualization After Cleaning

After cleaning by splitting the Intake column into four separate fields, three distinct analyses became possible that were previously unachievable with the raw data. Through the splitting process, four separate analytical columns were derived (Course, Education Level, Season, and Month Year), enabling the visualizations that follow.

Chart 1: Student Intake Trend by Month & Year

Purpose: To identify periods with the highest student application volumes.

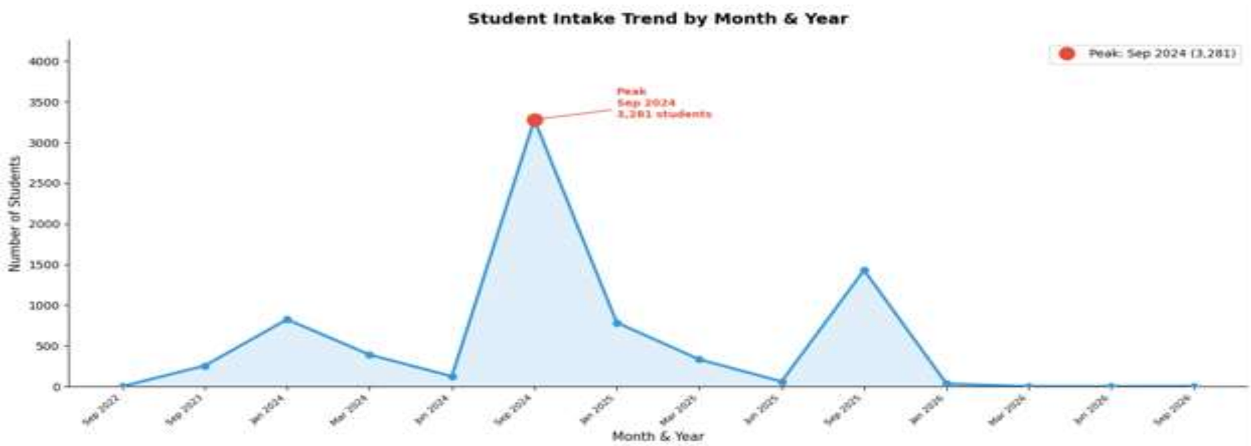

Figure 1.3 Student Intake Trend by Month & Year

Insight: The highest number of applications was recorded during the Fall 2024 intake period. Intake fluctuated over time with September 2024 recording the highest peak of 3,281 students, indicating strong student preference for fall admissions.

Chart 2: Student Intake by Season

Purpose: To analyze the distribution of applications across different seasons.

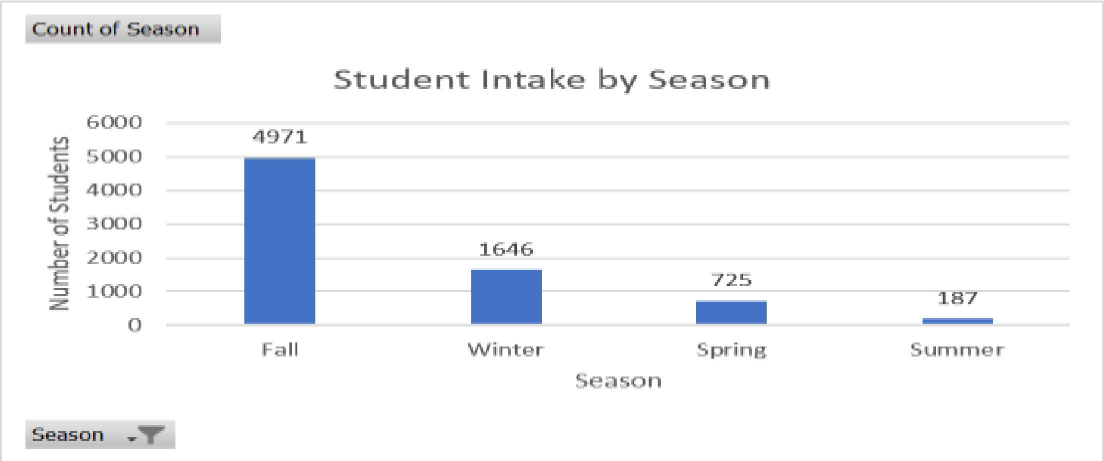

Figure 1.4: Student Intake by Season

Insight:

- Fall: 4,971 students (66%)
- Winter: 1,646 students (21.9%)
- Spring: 725 students (9.6%)
- Summer: 187 students (2.5%)

Fall admissions dominate the dataset and represent nearly two-thirds of all student applications. Summer intake is critically underutilized at just 2.5%, signalling either low student awareness of summer admission cycles or structural access barriers worth investigating.

Chart 3: Top 10 Most Applied Courses

Purpose: To identify the most popular programs among applicants

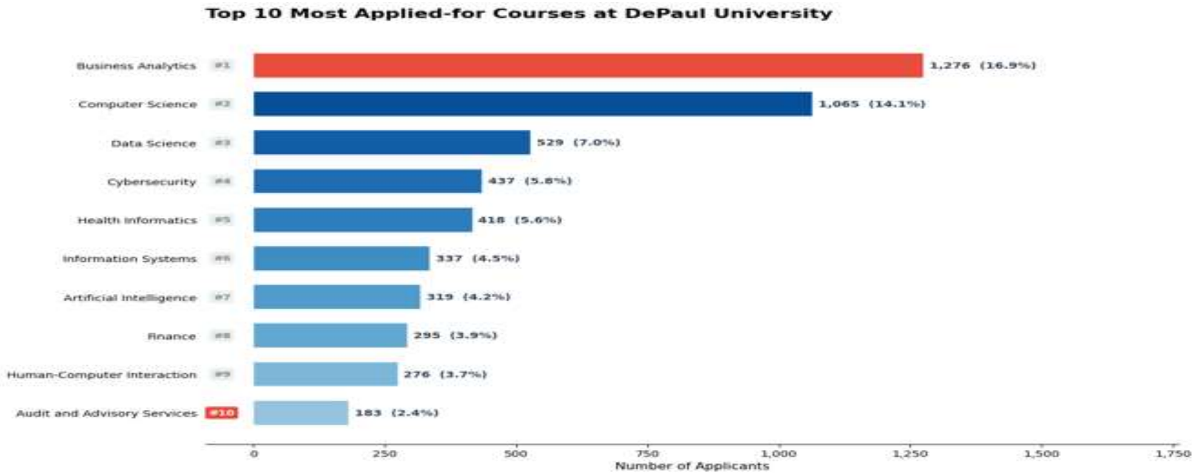

Figure 1.5: Top 10 Most Applied-for Courses

Insight: Business Analytics received the highest number of applications at 1,276 (16.9%), followed by Computer Science at 1,065 (14.1%) and Data Science at 529 (7.0%).

This reflects strong student demand for technology and data-driven disciplines and may inform resource allocation and program capacity decisions

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5. Short Summary Report

Impact of Data Cleaning on Dataset Reliability and Usability

The raw dataset contained 80 columns and 7,543 records, most of which were analytically unusable. 38 columns were completely empty, others held irrelevant system-generated data, and the Intake column stored multiple variables compressed into a single unstructured text string. After cleaning, the dataset was reduced to 8 columns and 7,529 records, with zero empty columns, no redundant attributes, and no unresolved missing values. The 14 records removed represented less than 0.2% of total data, a negligible loss.

The most significant transformation was engineering the Intake column into four separate fields: Course, Education Level, Season, and Month & Year. This single action converted an unanalyzable string into four independent dimensions, directly enabling the visualizations in this report.

Initial insights from the cleaned data:

- Fall 2024 dominates intake volume. September 2024 alone accounted for 3,281 students, nearly 44% of all records in the dataset.
- Business Analytics is the most applied-for course with 1,276 applicants (16.9%), followed by Computer Science at 1,065 (14.1%). Demand is heavily concentrated in technology and data fields.
- Summer intake is critically underutilized at just 2.5% of total applications. This is the most actionable anomaly in the dataset and warrants further investigation in subsequent analysis phases.
- The cleaned dataset is fully structured and ready to inform Week 3 dashboard metrics and validation logic.

6. Conclusion

The data cleaning process successfully transformed the raw dataset into a structured and analysis-ready format. Missing values were handled appropriately, irrelevant attributes were removed, and new features were generated to support meaningful analysis. The cleaned dataset provided clear insights into student intake trends, seasonal preferences, and popular academic programs.